HSAN ELLOUZE

Software Engineering Student | LLM Applications, Observability & Evaluation
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Seeking a 6-month end-of-studies internship (PFE) starting February 2027 | Open to relocation (Berlin)

Summary

Final-year Software Engineering student (INSAT, expected graduation September 2027) who has shipped LLM applications to production (RAG chatbots, LLM-backed contract analysis) and built multi-agent observability tooling on logs and metrics. Works across the LLM application lifecycle: retrieval, agent orchestration (LangChain/LangGraph), comparing output quality across model providers, automated testing and deployment. Fullstack TypeScript and Python background (Next.js, React, Express.js, FastAPI, Docker, CI/CD) to build developer-facing products end to end.

Experience

AI Engineer Intern – Applicare, Canada Jun 2026 – Aug 2026
- Architected AskSahera, a bilingual (FR/EN) **production LLM application** (RAG chatbot) indexing ∼100 source documents
- Built the RAG backend in Express.js with Qdrant vector storage and OpenAI embeddings, using Claude for generation; integrated Cloudflare R2 for storage
- Developed a dual-source ingestion pipeline (PubMed XML + GROBID PDF parsing) with IMRAD-aware chunking and a sliding-window fallback

Mazdev – Tunis
Software Engineer Intern Jul 2025 – Sep 2025
- Created a RAG engine (Pinecone) reaching 94% semantic accuracy across a large document set; optimized embeddings to cut **latency** 45%
- Established an **automated testing** framework for the chatbot platform, cutting the QA cycle 50%
- Engineered a production chatbot platform (YAML-configurable) used by non-technical users to create 20+ bot variants

Fullstack Developer Mar 2025 – Jun 2025
- Architected an AI-powered legal contract review platform for 40+ enterprise clients, cutting analysis time 60%
- Delivered a React/Next.js + FastAPI RESTful API system processing 500+ contract reviews per month at production scale
- Implemented a Docker CI/CD pipeline, cutting infra setup time 75%

Frontend Developer Intern – Steros, Sfax Jul 2024 – Aug 2024
- Built a responsive showcase website and interactive UI components, with a PHP/MySQL backend

Projects

KubeMind: AI-Driven Kubernetes Observability Platform (3-person capstone team) GitHub
- Implemented 3 specialized LangChain/LangGraph agents (logs, metrics, events) orchestrated via a LangGraph StateGraph workflow, feeding a correlation agent
- Evaluated **LLM output quality** across 3 providers (OpenAI, Gemini, Ollama) on 3 controlled incident scenarios
- Developed a FastAPI backend and Next.js dashboard on a real-time **observability** ingestion pipeline (Prometheus, Loki, Kafka, MongoDB)
- Designed a 5-layer agentic software architecture (ingestion, AI agents, correlation/storage, presentation) for automated incident diagnosis

Multi-Tenant AI Chatbot Widget GitHub
- Designed a reusable, embeddable AI chatbot component for multi-tenant deployment across client websites via a single `<script>` tag
- Enforced Shadow DOM isolation and strict domain validation for secure multi-tenant use
- Integrated the backend with Hono, Supabase, and Upstash Redis; deployed on Railway and Cloudflare Pages

PromptCraft: AI NFT Generation & Auction Platform GitHub
- Created an end-to-end platform combining AI-generated artwork with a live auction marketplace
- Engineered a WebSocket/SSE bidding system supporting 100+ concurrent users, cutting latency ∼60%

Education

INSAT – National Institute of Applied Sciences and Technology Sep 2022 – Sep 2027 (expected)
Engineering Degree in Software Engineering (EQF Level 7 / Master's equivalent)

Skills

AI/ML: LLM Integration, RAG, LangChain, LangGraph, Prompt Engineering, Embeddings, Vector Databases (Qdrant, Pinecone)
Languages: TypeScript, Python, JavaScript, SQL, Java, C#, PHP
Frontend: Next.js, React, Vue
Backend: Express.js (JavaScript), Hono (TypeScript), NestJS (JavaScript), FastAPI (Python), ASP.NET Core (C#)
Data & Observability: Redis (Upstash), Prometheus, Loki, Kafka, MongoDB, MySQL, Supabase
DevOps & Cloud: Docker, Kubernetes, GitHub Actions, CI/CD, Vercel, Railway, Cloudflare (Pages, R2)
Tools: Git, Linux

Activities

Event Mission Leader – Junior Enterprise INSAT 2024 – 2025
- Supervised a ∼40-member team over the 2024–2025 mandate, leading the organizing committee for 2 flagship events, including Forum INSAT Entreprise (1,500+ attendees, 40+ companies)

Operations Manager – WinterCup 7.0 2025
- Owned end-to-end logistics and event planning for WinterCup 7.0, an overnight coding contest with ∼300 participants, coordinating the on-site operations staff under time pressure